

H2 CHEMISTRY 9647

Paper 1 Multiple Choice Questions Worked Solutions

- 1 Since 6.02×10^{23} of particles is the same as one mole of the particles
 $\Rightarrow$ just need to determine the number of moles of the stated particles in each case

A: $n_e = n_{H_2} \times 2 = \frac{m}{M} \times 2 = \frac{1}{2.0} \times 2 = 1.0 \text{ mol}$

B: $n_{CH_3CO_2H} = c \times V = 1 \times 1 = 1.0 \text{ mol}$

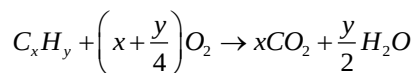
However, since ethanoic acid is a weak acid, $n_{H^+} \ll n_{CH_3CO_2H}$

C: $n_{Cl_2} = \frac{m}{M} = \frac{35.5}{71.0} = 0.5 \text{ mol}$

D: $n_{O \text{ atoms}} = n_{O_2} \times 2 = \frac{V}{V_m} \times 2 = \frac{22.4}{22.4} \times 2 = 2.0 \text{ mol}$

Ans: A

- 2 Let the formula of the hydrocarbon be C_xH_y , the equation of the combustion is therefore:



Since the volume of gaseous mixture decreased by 30 cm^3 when it was passed through $NaOH(aq) \Rightarrow V_{CO_2} = 30 \text{ cm}^3$

Comparing the moles ratio to the volume ratio of C_xH_y with that of CO_2 :

$$\Rightarrow \frac{n_{CO_2}}{n_{C_xH_y}} = \frac{V_{CO_2}}{V_{C_xH_y}}, \Rightarrow \frac{x}{1} = \frac{30}{10}, \Rightarrow x = 3$$

Since after the combustion the volume of the mixture decreased by 25 cm^3 ,

$$\Rightarrow (V_{C_xH_y} + V_{O_2 \text{ reacted}}) - (V_{CO_2} + V_{H_2O}) = 25$$

Substituting the known volumes into the equation and taking volume of H_2O at r.t.p to be 0 cm^3 ,

$$\Rightarrow (10 + V_{O_2 \text{ reacted}}) - (30 + 0) = 25$$

$$\Rightarrow V_{O_2 \text{ reacted}} = 25 + 20 = 45 \text{ cm}^3$$

Comparing the moles ratio to the volume ratio of C_xH_y with that of O_2 reacted, and substituting $x = 3$ into the equation:

$$\Rightarrow \frac{n_{O_2 \text{ reacted}}}{n_{C_xH_y}} = \frac{V_{O_2 \text{ reacted}}}{V_{C_xH_y}}, \Rightarrow \frac{x + \frac{y}{4}}{1} = \frac{45}{10},$$

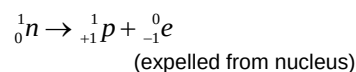
$$\Rightarrow 3 + \frac{y}{4} = 4.5, \Rightarrow \frac{y}{4} = 4.5 - 3 = 1.5$$

$$\Rightarrow y = 1.5 \times 4 = 6$$

Hence, the formula of the hydrocarbon is C_3H_6 .

Ans: B

- 3 The equation for the radioactive decay described in the question is:



(where n = neutron, p = proton, and e = electron)

Therefore, in this form of radioactive decay, the mass number of the nuclei will remain the same, while the proton number will increase by one unit.

A: ${}_{6}^{11}C \rightarrow {}_{6}^{12}C$, is not a process of this sort

B: ${}_{53}^{111}I \rightarrow {}_{52}^{111}Te$, is not a process of this sort

C: ${}_{35}^{76}Br \rightarrow {}_{36}^{76}Kr$, is a process of this sort

D: ${}_{35}^{76}Br \rightarrow {}_{35}^{75}Br$, is not a process of this sort

Ans: C

- 4 Element **E** must be a metal since it has a high melting point and is a good conductor of electricity (the only exception is probably graphite). The chloride of **E** must be a covalent compound as it does not conduct electricity in both the solid and molten state, and is soluble in a non-polar solvent like benzene. (Note that an ionic compound will be able to conduct electricity in the molten state, and will be insoluble in benzene).

A: Be is a metal, and $BeCl_2$ is a simple covalent compound (which has a polymeric structure, and that accounts for a melting point that is higher than normally expected of a simple covalent compound).

B: Si is a non-metal (and it is only a semi-conductor of electricity), and $SiCl_4$ is a simple covalent compound.

C: Cu is a metal, and $CuCl_2$ is an ionic compound.

D: Li is a metal, and $LiCl$ is an ionic compound.

Ans: A

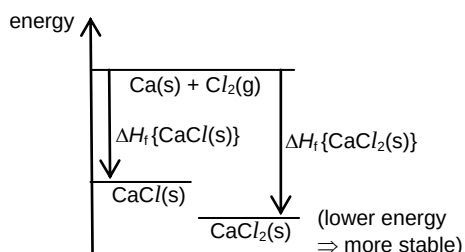
- 5 The structure of potassium chloride is the same as that of sodium chloride, where the cations and anions are arranged alternately in a regular three-dimensional array, where each K^+ ion is surrounded by six Cl^- ions, and each Cl^- ion is surrounded by six K^+ ions.

Options A, C, and D all have ions of the same charge placed side-by-side in the ionic lattice.

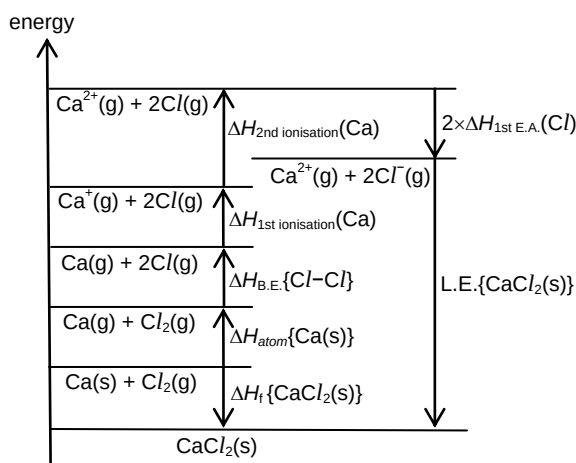
Ans: B

- 6 When a metal reacts with a non-metal, the ionic compound formed is always the one that is more stable / that has lower energy.

The stability of $\text{CaCl}_2(\text{s})$ can be compared with $\text{CaCl}(\text{s})$ using the enthalpy change of formation, ΔH_f , values. We know that $\text{CaCl}_2(\text{s})$ is more stable because ΔH_f for $\text{CaCl}_2(\text{s})$ is more exothermic.

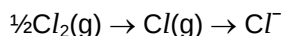


The value of $\Delta H_f\{\text{CaCl}_2(\text{s})\}$ depends on the various processes in the Born-Haber cycle:



It can be seen that ΔH_f will be more exothermic if the magnitude of the endothermic processes (atomisation of the elements and the ionisation of Ca) are smaller, and the magnitude of the exothermic processes (electron affinity of Cl and lattice energy) are greater.

- A: It is not possible to determine whether energy is released when chloride ions are formed from chlorine molecules, as the enthalpy change depends on the relative magnitude of the bond dissociation energy of the Cl-Cl bond (endothermic) and the first electron affinity of chlorine (exothermic).



Therefore, it is not possible to determine whether this statement is true or not.

- B: True statement.

Since less energy is required to ionise $\text{Ca}(\text{g})$ to form $\text{Ca}^+(\text{g})$, we would expect $\Delta H_f\{\text{CaCl}(\text{s})\}$ to be more exothermic than that of $\Delta H_f\{\text{CaCl}_2(\text{s})\}$, and so $\text{CaCl}(\text{s})$ will be formed instead of $\text{CaCl}_2(\text{s})$, which is not the case.

- C: False statement.

More energy is released when $\text{CaCl}_2(\text{s})$ is formed from its elements, and that is the reason why $\text{CaCl}_2(\text{s})$ will be formed instead of $\text{CaCl}(\text{s})$.

- D: True statement.

Since $|L.E.|\propto \left| \frac{q_+ \times q_-}{r_+ + r_-} \right|$, the lattice energy of

$\text{CaCl}_2(\text{s})$ is more exothermic than that of $\text{CaCl}(\text{s})$, as Ca^{2+} has double the charge than Ca^+ (and is smaller). Hence $\Delta H_f\{\text{CaCl}_2(\text{s})\}$ would be more exothermic than that of $\Delta H_f\{\text{CaCl}(\text{s})\}$, and so $\text{CaCl}_2(\text{s})$ will be formed instead of $\text{CaCl}(\text{s})$.

Ans: D

- 7 A: Correct

E° values should be measured under standard conditions of 298 K, 1.0 atm pressure, and using solutions containing 1.0 mol dm^{-3} of the aqueous ions.

- B: Correct

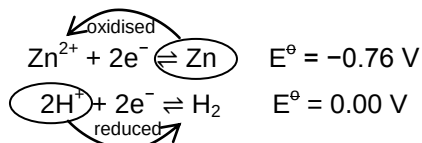
The cell e.m.f. should be measured using a high resistance voltmeter, so as to ensure that the electrical current flowing in the circuit is essentially zero, and that the voltmeter measures the maximum potential difference.

- C: Not correct

For the standard hydrogen electrode, $[\text{H}^+]$ should be 1.0 mol dm^{-3} . In a solution containing 1.0 mol dm^{-3} of H_2SO_4 , the $[\text{H}^+]$ is 2.0 mol dm^{-3} .

- D: Correct

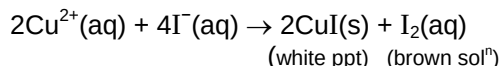
By comparing the E° values of the two half-reactions involved, it can be deduced that Zn will be oxidised to Zn^{2+} in Cell 1.



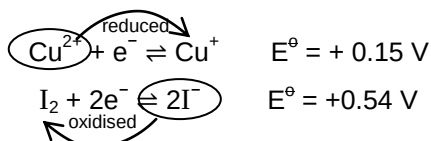
Therefore the Zn electrode will be the anode (negatively charged) and so electrons will flow from Cell 1 to the standard hydrogen electrode.

Ans: D

- 8 The equation for the reaction between CuSO_4 and KI is:



The two relevant half-cell reactions are:



$$\begin{aligned} E^{\circ}_{\text{cell}} &= +0.15 - (+0.54) \\ &= -0.39 \text{ V} \end{aligned}$$

Ans: D

- 9 From either of the graphs, it can be seen that as pressure increases (at constant temperature), the % yield of C (product) increases.

⇒ Equilibrium position shifts to the right to decrease pressure (by reducing the number of moles of gas).

⇒ Number of moles of gaseous products < number of moles of gaseous reactants.

⇒ $c < a + b$

From both graphs, it can be seen that when temperature changes from T_1 to T_2 (at the same pressure), % yield of C (product) increases (i.e. equilibrium position shifted to the right).

⇒ Since the forward reaction is exothermic, equilibrium position shifted to the right to give off heat

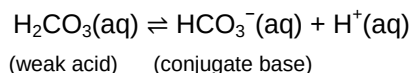
⇒ The disturbance introduced to the equilibrium must have been a decrease in temperature.

⇒ $T_2 < T_1$

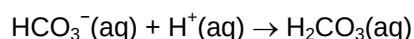
Ans: A

- 10 The major buffer system that is used to control the pH of blood is the carbonic acid, H_2CO_3 , and bicarbonate, HCO_3^{-} , buffer system.

The equilibrium reaction for the dissociation of carbonic acid, H_2CO_3 is given by:



When H^{+} is added, it is removed by the HCO_3^{-} present.



Ans: B

- 11 For enzyme-catalysed reactions, as there is a finite number of active sites that can be used to catalyse the reaction, the relationship between the rate of the reaction with [reactant] (the reactant in this question is ethanal) consist of two stages:

- At low concentration of [reactant], some of the active sites are still available, and so when [reactant] increase (slightly), more of the active sites get filled up, and the reaction becomes faster ⇒ rate is proportional to [reactant] and so the [ethanal] versus rate graph is an increasing straight line with a constant gradient.
- At high concentration of [reactant], all the active sites are filled up, and so when [reactant] increase, the rate remains the same ⇒ rate is independent of [reactant] and so the [ethanal] versus rate graph is a vertical line.

Ans: D

- 12 As the temperature increases from 273 K to 473 K, the Maxwell Boltzmann distribution curve will have:

- A broader base, as more particles have higher kinetic energy.
- A lower peak, as the total number of particles, which is proportional to the area under the curve, must remain the same.

Ans: C

- 13 Mg and S are the only two elements in Period 3 that have 1st I.E. larger than both the element before and after it. Obviously, Mg is the element that the question is referring to, as it is a metal (can conduct electricity), and can react with steam (but not with cold water) to form $\text{MgO}(\text{s})$ and $\text{H}_2(\text{g})$.

A: Correct

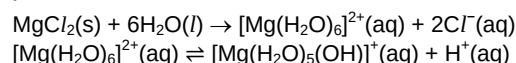
Mg has a giant metallic lattice, and so a large amount of energy is required to overcome the strong electrostatic attraction between the metallic ions and the 'sea' of delocalised electrons
⇒ it has high melting and boiling point.

B: Correct

The oxidation state of Mg in its compounds is usually +2, which is the same as its group number in the *Periodic Table*.

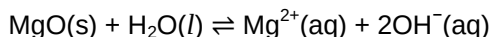
C: Correct

MgCl_2 produces a solution that is slightly acidic when dissolved in water, as $\text{Mg}^{2+}(\text{aq})$ undergo cation hydrolysis (slightly) to produce H^{+} ions:



D: Not correct

MgO is only slightly soluble in water (due to its high lattice energy), to form a slightly basic solution (rather than an acidic solution).

**Ans: D**

- 14**
- Na_2O can dissolve in / reacts with water to form NaOH(aq) :

$$\text{Na}_2\text{O(s)} + \text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)}$$
 NaOH(aq) does not react with $\text{Na}_2\text{CO}_3(\text{aq})$.
 $\Rightarrow \text{Na}_2\text{O}$ is **X**.
 - Al_2O_3 does not dissolve in water, due to its very high lattice energy.
 Al_2O_3 is an amphoteric oxide and undergo acid-base reaction with HCl(aq) to form AlCl_3 .

$$\text{Al}_2\text{O}_3(\text{s}) + 6\text{HCl(aq)} \rightarrow 2\text{AlCl}_3(\text{aq}) + 3\text{H}_2\text{O(l)}$$
 $\Rightarrow \text{Al}_2\text{O}_3$ is **Y**.
 - SiO_2 does not dissolve in water, due to the large amount of energy required to break the numerous covalent bonds in the giant covalent structure.
 SiO_2 is an acidic oxide and does not react with or dissolve in HCl(aq) .
 $\Rightarrow \text{SiO}_2$ is **Z**.
 - P_4O_{10} reacts with water to form $\text{H}_3\text{PO}_4(\text{aq})$:

$$\text{P}_4\text{O}_{10}(\text{s}) + 6\text{H}_2\text{O(l)} \rightarrow 4\text{H}_3\text{PO}_4(\text{aq})$$
 $\text{H}_3\text{PO}_4(\text{aq})$ is a strong acid, and it can react with $\text{Na}_2\text{CO}_3(\text{aq})$ to form CO_2 gas (that forms a white ppt with limewater).

$$2\text{H}_3\text{PO}_4(\text{aq}) + 3\text{Na}_2\text{CO}_3(\text{aq}) \rightarrow 2\text{Na}_3\text{PO}_4(\text{aq}) + 3\text{CO}_2(\text{g}) + 3\text{H}_2\text{O(l)}$$
 $\Rightarrow \text{P}_4\text{O}_{10}$ is **W**.

Ans: B

- 15** A transition element is defined as a d-block element that forms at least one or more stable ions with a partially filled d-subshell.

Ans: C

- 16** From the information given in the table, it can be deduced that:

- Complex I have three Cl^{-} ions, of which only one of them is bonded to the Co^{3+} ion, and the remaining two Cl^{-} ions are free to react with Ag^{+} to form AgCl . However, for complex II and III, two of the Cl^{-} ions are bonded to the Co^{3+} ion, and only one

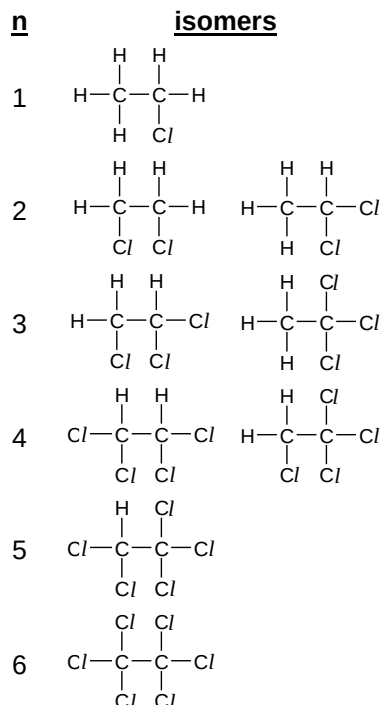
remaining Cl^{-} ions is free to react with Ag^{+} to form AgCl .

$\Rightarrow$ Either option C or D.

- Complex I have a resultant dipole moment, as there is only one Cl^{-} ion bonded to Co^{3+} in the complex, and the dipole moment of the Co-Cl bond and the Co-N bond do not cancel out completely.
 For complex II, as it has two Cl^{-} ions bonded to Co^{3+} in the complex and have a resultant dipole moment $\Rightarrow$ the two Cl^{-} ions must occupy adjacent positions in the octahedral arrangement (i.e. not at opposite sides of the octahedron), so that the dipole moments do not cancel out completely.
 For complex III, as it has two Cl^{-} ions bonded to Co^{3+} in the complex and have no resultant dipole moment $\Rightarrow$ the two Cl^{-} ions must occupy opposite positions in the octahedral arrangement, so that the dipole moments of the Co-Cl bond are cancelled out by the Co-Cl bond on the opposite side (and the dipole moments of the Co-N bonds are also cancelled out by each other).
 $\Rightarrow$ Option C.

Ans: C

- 17** The possible isomers with the formula $\text{C}_2\text{H}_{6-n}\text{Cl}_n$ are:

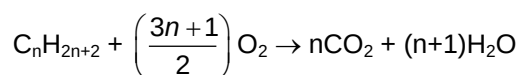


$\Rightarrow$ total number of isomers = 9

(Note that there are no stereoisomers possible)

Ans: C

- 18** General formula of aliphatic alkanes: C_nH_{2n+2} , and the balanced equation for the combustion of alkanes is:

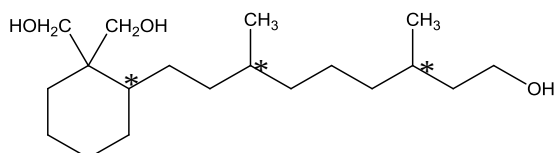


Hence for every increase of one carbon atom in the alkane, the number of moles of oxygen needed for the combustion will increase by $1\frac{1}{2}$ moles (e.g. CH_4 requires 2 moles of O_2 , while C_2H_6 requires $3\frac{1}{2}$ moles of O_2 , and C_3H_8 requires 5 moles of O_2 and so on)

⇒ There is a regular increase in the number of moles of O_2 required for the combustion, and this is represented by an increasing straight line graph

Ans: B

- 19** The product formed when the Vitamin A derivative reacts with excess $H_2(g)$ with Pt is:



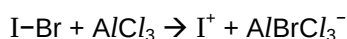
There are 3 chiral centres (indicated by *),
Total no. of stereoisomers $= 2^3 = 8$

Ans: B

- 20** **A:** As the $-CH_3$ group is 2,4-directing, the first step will produce either 1-methyl-4-nitrobenzene or 1-methyl-2-nitrobenzene, but not 1-methyl-3-nitrobenzene (which will lead to the desired product)
- B:** $Cr_2O_7^{2-}/H^+$ in the first step cannot be used to oxidise methylbenzene to benzoic acid
- C:** As the $-CN$ group is 3-directing, the substitution of the $-NO_2$ group will occur at the 3-position leading to the formation of the desired product
- D:** $NaBH_4$ in the third step cannot be used to reduce $-NO_2$ group to $-NH_2$ group to form the desired product

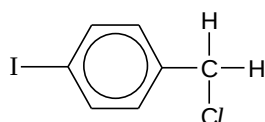
Ans: C

- 21** Since I is less electronegative than Br, the δ^+ charge will reside on the I-atom in the I-Br molecule. The molecule will therefore react with $AlCl_3$ to produce the electrophile I^+ :

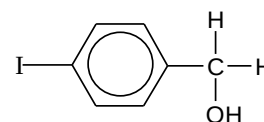


I^+ then acts as the electrophile to attack the benzene ring.

Structure of **T** is therefore:



Since halogenoarenes is unreactive, only the Cl atom will be hydrolysed by $NaOH(aq)$ under reflux. Hence, the final product obtained is:



Ans: B

- 22** The mechanism for the hydrolysis of 2-bromobutane given in the question is S_N2 , which consists of only one step. Therefore the reaction profile diagram should only have one "hump" (the reactants have to overcome an energy barrier, E_a , to form the products, which is required to break the C-Br covalent bond).

Ans: D

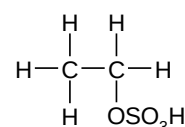
- 23** Dichlorodifluoromethane is chemically inert as the C-F bonds are very strong (bond energy $= 440 \text{ kJ mol}^{-1}$).

As fluorine is very electronegative, the δ^+ charge on the carbon atom is quite large (i.e. the C-F bond is very polar), and so the C-atom should be very susceptible to attack by nucleophiles, and so dichlorodifluoromethane should be quite reactive (which is not the case).

Ans: A

- 24** In the experimental setup, one reagent is added (from a dropping funnel) to another in an ice water bath, i.e. the reaction is carried out in cold conditions.

- A:** The reaction between phenol and Br_2 in CCl_4 in the cold will ensure that monosubstitution takes place. The cold condition prevents further substitution of the phenol.
- B:** The reagents and conditions for the conversion of ethene to ethanol should be: cold, concentrated H_2SO_4 , followed by H_2O , heat. The reaction between ethene and cold concentrated H_2SO_4 will only produce the following compound:



- C:** The hydrolysis of 2-chloropropanone with $NaOH(aq)$ to form propan-2-ol requires heating (or reflux conditions).
- D:** The formation of ester from alcohol and carboxylic acid in the presence of c. H_2SO_4 catalyst requires heating (or reflux conditions).

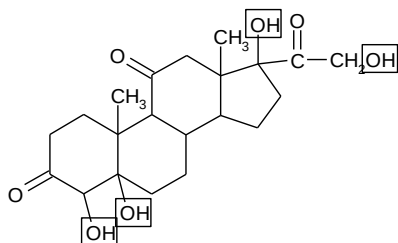
Ans: A

- 25** **A:** Brady's reagent (2,4-DNPH) will give positive test for both aldehydes and ketones.
- B:** Fehling's reagent only gives brick red ppt of Cu_2O for aliphatic aldehyde. Hence, it will give negative test for both compounds.
- C:** Na gives effervescence of $\text{H}_2(\text{g})$ when added to alcohol, phenol or carboxylic acid. Hence, it will give negative test for both compounds.
- D:** Aqueous alkaline iodine will give a yellow ppt of CHI_3 for 1-phenylethanone but no ppt for benzaldehyde.

Ans: D

- 26** **A: False**

Only the alkene functional group in cortisone will undergo mild oxidation with cold dilute KMnO_4 to form a product with 4 hydroxyl (alcohol) groups.

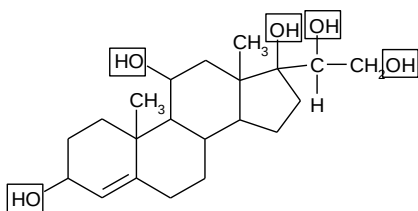


- B: False**

Cortisone does not contain the $-\text{COCH}_3$ or $-\text{CH}(\text{OH})\text{CH}_3$ structure

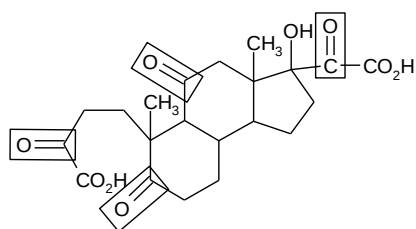
- C: True**

The three ketone functional groups in cortisone will undergo reduction with NaBH_4 (methanol is the solvent) to form a product with 5 hydroxyl (alcohol) groups.



- D: False**

The primary alcohol functional group in cortisone will be oxidised to a carboxylic acid, while the alkene functional group will undergo oxidative cleavage with hot acidified KMnO_4 to form a product with 4 carbonyl groups

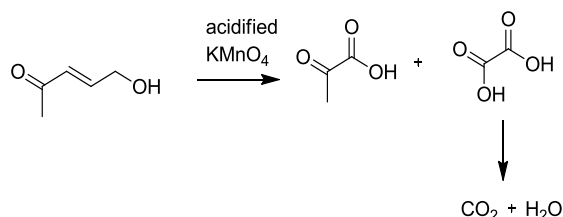


(Note that the tertiary alcohol cannot be oxidised, and the $\text{C}=\text{O}$ in carboxylic acids are not carbonyl groups).

Ans: C

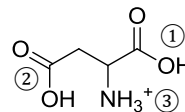
- 27** Compound **S** produces an orange precipitate when added to 2,4 dinitrophenylhydrazine but no precipitate is observed when warmed with Tollen's reagent. Hence, it must be a ketone. Only option A and C have ketone functional group.

When heated with dilute acidified potassium manganate(VII), only option A produces ethanedioic acid which can be further oxidised to carbon dioxide gas (whereas option C will produce ethanoic acid).



Ans: A

- 28** The fully protonated form of aspartic acid is a tribasic acid:



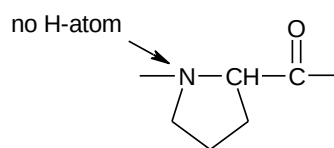
Deprotonation occurs when the given pH is above the pK_a value. At pH 3, deprotonation only occurs at ① ($\alpha\text{-CO}_2\text{H}$) where $\text{pK}_a = 2.10$. No deprotonation occurs at ② ($-\text{CO}_2\text{H}$ at side chain) where $\text{pK}_a = 3.86$ and at ③ ($\alpha\text{-amino}$ group) where $\text{pK}_a = 9.82$.

Ans: C

- 29** The type of reaction that occurs when a protein is broken down into peptides and amino acids is hydrolysis

Ans: A

- 30** Proline (Option D) contains a secondary amine group. After forming peptide linkage, proline has no more N-H group available to form hydrogen bond to stabilise the peptide α -helix.



Ans: D

31 1: Correct

At constant T, average K.E. remains constant

2: Correct

$pV = nRT \Rightarrow M_r = mRT/pV$
 Since p, V and T are constant,
 $M_r(S) : M_r(T) = m(S) : m(T)$
 $= 0.5 \text{ g} : 2.5 \text{ g}$
 $= 1 : 5$

3: Wrong

Since $n = pV/RT$ and p, V and T are constant, n is constant and hence no. of molecules is constant for X and Y.

Ans: B**32 1: Correct**

Dissolving the solute in the solvent
 Increases the number of ways of arranging molecules in the solution which will lead to an increase in entropy (positive ΔS).

2: Correct

Evaporation causes the solvent to change from liquid to gas. This change in state increases the disorderliness in the system (positive ΔS).

3: Correct

The movement of solvent molecules from the pure solvent through the semi-permeable membrane to the solution during osmosis increases the number of ways of arranging solvent molecules in the solution which will lead to an increase in entropy (positive ΔS).

Ans: A**33** n_{e^-} transferred = $290/96500 = 3.01 \times 10^{-3} \text{ mol}$

$$\frac{n_{e^-}}{n_M} = \frac{3.01 \times 10^{-3}}{1.00 \times 10^{-3}} = 3$$

Hence, at the cathode, $M^{x+} + 3e^- \rightarrow M$
 Each mole of metal cation requires 3 mol of electrons for reduction, hence $x = 3$ and cation has a charge of +3.
 Only iron and chromium can form Fe^{3+} and Cr^{3+} respectively.

Ans: C**34 1: Correct**

Recall that for clock experiment, Rate $\propto 1/t$, and letting the rate equation be:

$$\text{rate} = k [L]^a [M]^b [N]^c$$

- Comparing expt 1 & 2, no change in rate even though [L] is varied
 $\Rightarrow$ zero order wrt L.

- Comparing experiment 2 & 3,

$$\frac{\text{rate}_2}{\text{rate}_3} = \frac{k[M]_2^b [N]_2^c}{k[M]_3^b [N]_3^c} = \frac{0.05^b}{0.10^b} = \frac{1/176}{1/44}$$

$$\Rightarrow b = 2$$

- Comparing experiment 3 & 4,

$$\frac{\text{rate}_3}{\text{rate}_4} = \frac{[N]_3^c}{[N]_4^c} = \frac{0.05^c}{0.10^c} = \frac{1/44}{1/22}$$

$$\Rightarrow c = 1$$

Hence, rate = $k[M]^2[N]$

2: Correct

Rearranging the rate equation:

$$k = \frac{\text{rate}}{[M]^2[N]} \cdot \frac{\text{mol dm}^{-3} \text{s}^{-1}}{(\text{mol dm}^{-3})^3} = \text{mol}^{-2} \text{dm}^6 \text{s}^{-1}$$

3: Wrong

Rate = $k[M]^2[N] = k'[N]$ where $k' = k[M]^2$
 $t_{1/2} = \ln 2 / k' = \ln 2 / k[M]^2$
 Hence $t_{1/2}$ of reaction is dependent on $[M]$.

Ans: B**35 1: Correct**

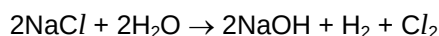
ΔH_{hyd} of cation $\propto q^+/r_+$
 Since ionic radius of M^{2+} increases down the group, ΔH_{hyd} of M^{2+} becomes less exothermic (decreases in magnitude).

2: Wrong

Reducing power of M $\Rightarrow$ Ease of removal of electrons. Since atomic radius of M increases down the group, valence electrons are more easily removed
 $\Rightarrow$ Down the group the elements are stronger reducing agents

3: Wrong

The atomic radius of the elements increases down the group due to the increase in number of principal quantum shell of electrons (the increase in screening effect outweighs the increase in nuclear charge)

Ans: D**36** The overall equation for the electrolysis of brine is as follows (by combining the half-reactions at cathode and anode):

If the products are allowed to react in the cold, NaOH and Cl_2 will react to form NaClO:



Hence the final products are NaCl, NaClO and H_2 .

Ans: B**37 1: Correct**

The electron rich nucleophile, OH^- substituted the Br-atom in the bromoalkane.

2: Wrong

The type of reaction is reduction

3: Correct

The type of reaction is electrophilic substitution

Ans: D

38 1: Correct

Two ions (formed by the reaction of ethanol and sulfuric acid in step 1) are reacting in the slow step which corresponds to 2nd order reaction.

2: Correct

H₂SO₄ is used up in the first step, and regenerated in the second step of the mechanism - the characteristic of a catalyst

3: Wrong

2-methylpropan-2-ol is a tertiary alcohol and is more likely to undergo dehydration via an elimination mechanism which is similar to S_N1 where a water molecule is lost to generate a stable tertiary carbocation. The slow step will only involve one molecule.

Ans: B**39 1: Wrong**

Phenol does not react with PCl₅(s) to give white fume of HCl due to partial double bond character of C-O bond in phenol.

2: Correct

Ethanol reacts with Na(s) to give H₂ gas

3: Correct

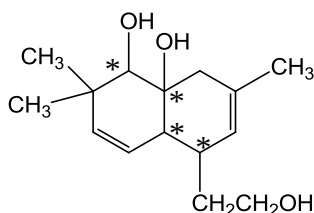
Ethanoic acid reacts with Na₂CO₃(aq) to give CO₂ gas

Ans: C**40 1: Correct**

Only carboxylic acid in compound **U** will react with NaOH(aq). Hence, 1 mol of **U** (which has only one -COOH group) will react exactly with 1 mol of NaOH(aq).

2: Correct

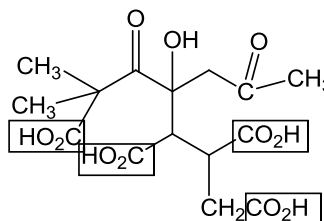
LiAlH₄ will reduce both ketone and carboxylic acid in compound **A** to produce a compound with 4 chiral centres. This will give 2⁴ = 16 stereoisomers.



(Note that double bonds within a ring cannot exhibit cis-trans isomerism due to ring strain)

3: Correct

Oxidative cleavage of the two alkene functional groups occurs with hot acidified KMnO₄ to give a product having four -CO₂H groups

**Ans: A**

Paper 2 Structured Questions

Question 1

(a) $K_{sp} = [\text{Ca}^{2+}][\text{OH}^-]^2$; units: $\text{mol}^3 \text{dm}^{-9}$

(b)(i) percentage error = $\frac{2 \times 0.05}{4.50} \times 100\% = 2.22\%$

The percentage error in the measurement of the titre volume is **relatively significant / high**, as the volume of hydrochloric acid required to react with **FA1** is too small, as the concentration of hydrochloric acid in **FA1** is too high.

- (ii) In the trial experiment, the amount of hydrochloric acid required to react completely with 25.0 cm^3 of **FA1**, $n_{\text{HCl}} = 0.200 \times 4.50 \times 10^{-3} = 9.00 \times 10^{-4} \text{ mol}$

Since the concentration and volume of $\text{Ca}(\text{OH})_2$ used in the actual titration is the same as that in the preliminary test, the amount of hydrochloric acid needed to neutralise $\text{Ca}(\text{OH})_2$ is the same.

$\Rightarrow$ For a titre reading of about 30 cm^3 , $[\text{HCl}]_{\text{FA2}} = \frac{9.00 \times 10^{-4}}{30 \times 10^{-3}} = 0.0300 \text{ mol dm}^{-3}$

number of moles of hydrochloric acid in 250 cm^3 of **FA2**

$= 0.0300 \times 250 \times 10^{-3} = 7.50 \times 10^{-3} \text{ mol}$

volume of stock solution required for dilution = $\frac{7.50 \times 10^{-3}}{0.200} = 0.0375 \text{ dm}^3 = 37.5 \text{ cm}^3$

Or

Since the titre volume in the preliminary test is 4.5 cm^3 , for a titre volume of around 27 cm^3 ($= 4.5 \text{ cm}^3 \times 6$) in the actual titration, the concentration of hydrochloric acid in **FA2** needs to be one-sixth that of the stock solution.

$\Rightarrow [\text{HCl}]_{\text{FA2}} = \frac{1}{6} \times 0.200 = 0.0333 \text{ mol dm}^{-3}$.

$\Rightarrow$ volume of stock solution required for dilution = $\frac{1}{6} \times 250 = 41.67 \text{ cm}^3$

(c) **A. Preparation of FA2**

- Fill a **burette** with the stock solution of hydrochloric acid. Drain **37.50 cm^3** of the hydrochloric acid from the burette into a **250 cm^3 volumetric flask**. **Top-up** the volumetric flask up to the mark with **de-ionised water**. Add the last few drops of de-ionised water using a dropper till the bottom of the meniscus coincides with the mark.
- Stopper** the volumetric flask firmly and invert it 10 to 15 times **to ensure complete mixing**. This is to ensure that a homogeneous solution is obtained.

B. Titration of FA1 against FA2

- Fill a clean **burette** with **FA2** solution. **Pipette 25.0 cm^3** of **FA1** into a clean **conical flask**.
- Add **2 or 3 drops** of **methyl orange / bromothymol blue / cresol red indicator** and add around 25 cm^3 of the **FA2** solution from the burette quickly to the contents in the conical flask with constant swirling, and then drop-wise as the titration is near the end-point.
- Stop the titration immediately when the first drop of **FA2** solution added turns the solution from **yellow to orange / blue to green / reddish purple to yellow**.
- Repeat the titration** until two accurate titre readings fall within 0.10 cm^3 of each other are obtained

(d)(i) $n_{HCl} = M \times y \times 10^{-3} \text{ mol}$

$$n_{Ca(OH)_2} = \frac{1}{2} \times n_{HCl} = \frac{1}{2} \times M \times y \times 10^{-3} \text{ mol}$$

$$\Rightarrow [Ca(OH)_2]_{FA1} = \frac{\frac{1}{2} \times M \times y \times 10^{-3}}{25.0 \times 10^{-3}} = \frac{1}{50} My \text{ mol dm}^{-3}$$

(ii) Since $[Ca(OH)_2]_{FA1} = \frac{1}{50} My \text{ mol dm}^{-3}$,

$$\Rightarrow [Ca^{2+}] = \frac{1}{50} My \text{ mol dm}^{-3} \text{ and } [OH^-] = 2 \times \frac{1}{50} My = \frac{1}{25} My \text{ mol dm}^{-3}$$

$$\Rightarrow K_{sp} = [Ca^{2+}][OH^-]^2 = \left(\frac{1}{50} My\right) \left(\frac{1}{25} My\right)^2 = 3.2 \times 10^{-5} M^3 y^3 \text{ mol}^3 \text{ dm}^{-9}$$

(e) Both HCl and Ca(OH)₂ are **corrosive. Wear gloves / put on a laboratory coat / put on goggles / use a dropper** when handling the chemicals

Or

HCl is toxic. Use **pipette filler** when siphoning the required volume from the stock solution into the pipette.

Or

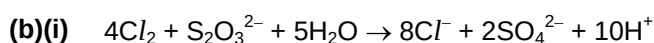
HCl is corrosive. Wear goggles (when filling the burette) to prevent accidental spillage into the eyes.

Question 2

Total Marks: 16, Average Mark: 6.8 (42.5%)

(a)

chemical parameters	average reading / mg dm ⁻³	average reading / mol dm ⁻³
Cl ₂ (aq)	2.3	3.24 × 10 ⁻⁵
NH ₂ Cl (aq)	2.0	3.88 × 10 ⁻⁵
PO ₄ ³⁻	0.012	1.26 × 10 ⁻⁷
Fe (as Fe ²⁺ & Fe ³⁺)	0.006	1.08 × 10 ⁻⁷



(ii) volume of tank water to be changed = $\frac{50}{100} \times 360 = 180 \text{ dm}^3$

number of moles Cl₂ present = $3.24 \times 10^{-5} \times 180 = 5.83 \times 10^{-3} \text{ mol}$

number of moles NH₂Cl present = $3.88 \times 10^{-5} \times 180 = 6.98 \times 10^{-3} \text{ mol}$

number of moles of S₂O₃²⁻ to react with Cl₂ = $\frac{1}{4} \times 5.83 \times 10^{-3} = 1.46 \times 10^{-3} \text{ mol}$

number of moles of S₂O₃²⁻ to react with NH₂Cl = $2 \times 6.98 \times 10^{-3} = 0.0140 \text{ mol}$

volume of S₂O₃²⁻ required = $\frac{1.46 \times 10^{-3} + 0.0140}{0.010} = \frac{0.0154}{0.010} = 1.54 \text{ dm}^3$

- (c) For precipitation to occur,
- $Q \geq K_{sp}$

$$[\text{Ca}^{2+}]^3 \times (1.263 \times 10^{-7})^2 \geq 2.07 \times 10^{-33}$$

$$\Rightarrow [\text{Ca}^{2+}] \geq 5.64 \times 10^{-7} \text{ mol dm}^{-3}$$

The minimum concentration of Ca^{2+} is $5.64 \times 10^{-7} \text{ mol dm}^{-3}$.

- (d)(i) Fe $1s^2 2s^2 2p^6 \underline{3s^2 3p^6 3d^6} 4s^2$
 Fe^{2+} $1s^2 2s^2 2p^6 \underline{3s^2 3p^6 3d^6}$

- (ii) In the presence of water ligands, the degenerate d-orbitals of $\text{Fe}^{2+} / \text{Fe}^{3+}$ are split into two sets of different energy levels.

When light is shone on the complex, an electron from the lower energy d-orbital of $\text{Fe}^{2+} / \text{Fe}^{3+}$ will absorb photons of wavelength / frequency in the visible light region and be promoted to the higher energy d-orbital.

Colours not absorbed (complementary colour) will form the observed colour of the complex.

- (e)(i) Cl_2 is a stronger oxidising agent than Br_2 and so it is able to displace Br^- from the solution to form Br_2 .

- (ii)

	chlorine, Cl_2	bromine, Br_2
in water	yellow / yellowish green / pale green	(lighter colour) yellow / orange
in hexane	colourless	orange / reddish-brown

- (iii) To provide heat energy to vaporise liquid Br_2 .

- (iv) H_2SO_4 has a higher boiling point than Br_2

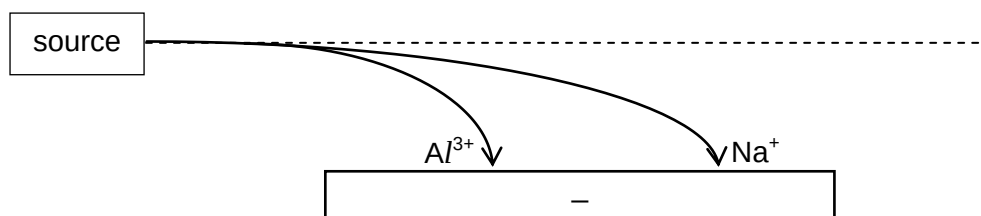
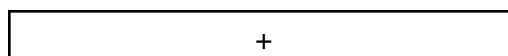
Or

H_2SO_4 is able to form stronger interactions with water via hydrogen bonding / ion-dipole interactions (H^+ and SO_4^{2-}) than Br_2 , and thus remains solvated in water.

Question 3 – Atomic Structure, Periodicity and Chemical Bonding

Total Marks: 15, Average Mark: 5.8 (38.7%)

- (a)

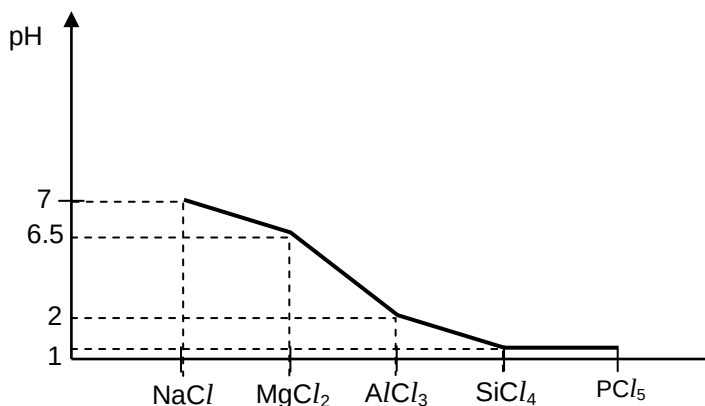


- (b)(i) Al and Al^+ have the **same nuclear charge** due to both having same number of protons but Al^+ has a **lower shielding effect** due to a **lower number of electrons**. **More energy** is required to overcome the **stronger net attractive force between the nucleus and valence electron of Al^+** than Al . Hence the 2nd IE of Al is higher than the 1st IE.

- (ii) Al^+ : $[Ne] 3s^2$
 Si^+ : $[Ne] 3s^2 3p^1$

The **3p electron in Si^+** is at a **higher energy level** than the **3s electron in Al^+** and thus experiences **weaker attraction to the nucleus**. **Less energy is required*** to remove the 3p electron in Si and 2nd IE of Si is lower than that of Al.

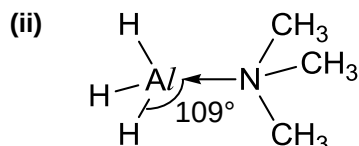
(c)(i)



- (ii) $AlCl_3 + 6H_2O \rightarrow [Al(H_2O)_6]^{3+} + 3Cl^-$
 $[Al(H_2O)_6]^{3+} \rightarrow [Al(H_2O)_5(OH)]^{2+} + H^+$ or $[Al(H_2O)_6]^{3+} \rightleftharpoons [Al(H_2O)_5(OH)]^{2+} + H^+$
 $SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$
 $PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$

- (d) Aluminium fluoride has a **giant ionic lattice structure** with **strong electrostatic forces of attraction between Al^{3+} and F^- ions** while aluminium chloride has a **simple molecular structure** with **weak instantaneous dipole-induced dipole interaction between the molecules**. **More energy** is required to overcome the **stronger** ionic bonds in aluminium fluoride, thus it has a higher melting point than aluminium chloride.

(e)(i) Lewis acid / catalyst



- (iii) The **lone pair of electrons on N** of $(CH_3)_3N$ was donated to the **empty 3p orbital of Al** in AlH_3 to form a **dative bond**.

Question 4 – Chemical Equilibrium, Hydrocarbons, Hydroxy compounds and Chemical Bonding

Total Marks: 14, Average Mark: 5.5 (39.3%)

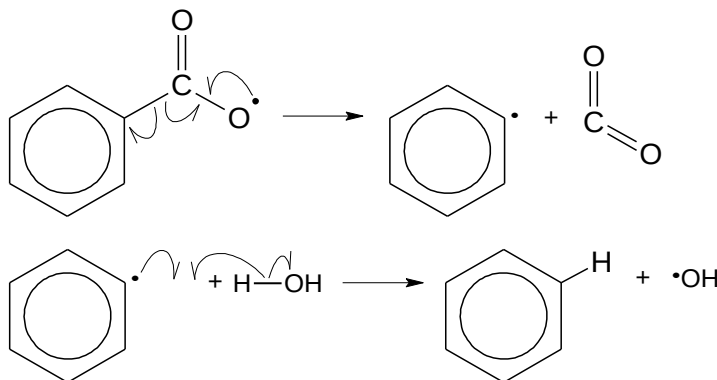
- (a)(i)
- | | $2HNO_3(aq)$ | $\rightleftharpoons NO_2^+(aq)$ | $+ NO_3^-(aq)$ | $+ H_2O(l)$ |
|---------------------------------|--------------|---------------------------------|--------------------|-------------|
| Initial (mol dm ⁻³) | y | 0 | 0 | |
| Change | -0.95y | $+\frac{0.95}{2}y$ | $+\frac{0.95}{2}y$ | |
| Eqm (mol dm ⁻³) | 0.05y | $\frac{0.95}{2}y$ | $\frac{0.95}{2}y$ | |

As given in the question, equilibrium concentration of NO_2^+ and NO_3^- are both 0.100 mol dm⁻³,
 $\Rightarrow \frac{0.95}{2}y = 0.100, \Rightarrow y = \frac{0.100 \times 2}{0.95} = 0.2105 \text{ mol dm}^{-3}$

(ii)
$$K_c = \frac{[\text{NO}_2^+][\text{NO}_3^-]}{[\text{HNO}_3]^2} = \frac{[0.1][0.1]}{[0.05 \times 0.2105]^2} = \frac{0.01}{0.0105^2} = 90.3 (\text{no units})$$

- (iii) NO_2^+ will undergo **electrophilic substitution** with the benzene ring (of phenol) to form nitrophenol. As a result **$[\text{NO}_2^+]$ will decrease**.
By Le Chatelier's principle, the position of equilibrium will shift to increase the $[\text{NO}_2^+]$. Hence, position of equilibrium will **shift right, favouring the forward reaction** or the **forward reaction will occur to a greater extent, increasing the yield of NO_3^-** .

(b)



- (c)(i) reagents: dilute HNO_3 / $\text{HNO}_3(\text{aq})$
conditions: room temperature

- (ii) **Intramolecular hydrogen bonding** exist between the $-\text{OH}$ group and $-\text{NO}_2$ of 2-nitrophenol, hence **2-nitrophenol** forms **less extensive intermolecular hydrogen bonding** with surrounding water molecules compared to 4-nitrophenol.
Lesser energy is released to overcome the hydrogen bonding between water molecules and the hydrogen bonding / van der Waals' forces of attraction between 2-nitrophenol molecules.
- (iii) Since the nitro group is electron-withdrawing (or the nitrogen atom is electronegative), the negative charge on the conjugate base of 4-nitrophenol is more dispersed compared to that of phenol
 $\Rightarrow$ the conjugate base of 4-nitrophenol is more stable than that of phenol
 $\Rightarrow$ 4-nitrophenol has a greater tendency to dissociate to produce H^+ ions
 $\Rightarrow$ 4-nitrophenol is a stronger acid than phenol
 $\Rightarrow$ $\text{p}K_a$ of 4-nitrophenol is lower than that of phenol.

Question 5 – Hydroxy Compounds, Carbonyl Compounds and Introduction to Organic Chemistry

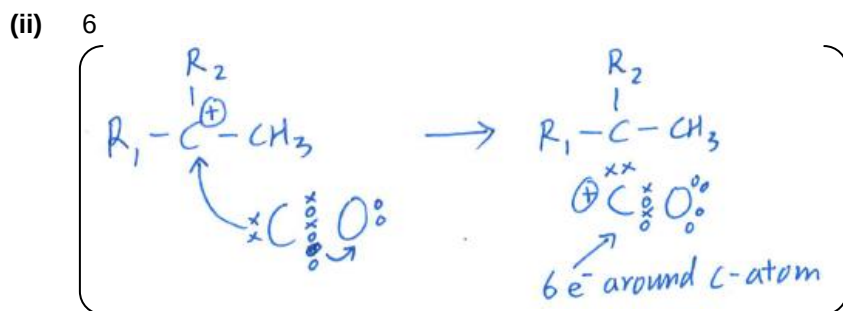
Total Marks: 15, Average Mark: 7.5 (50.0%)

(a)

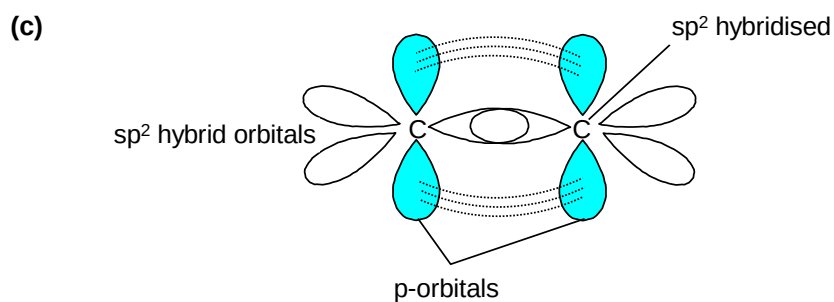
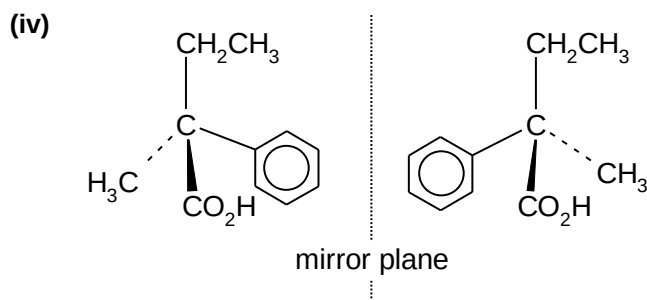
reagent	compound W, X, Y or Z	structural formula of the organic product(s)
PBr_3	W	
2,4-dinitro-phenylhydrazine	X	

Fehling's reagent, heat	X	
acidified $K_2Cr_2O_7(aq)$, heat	W	

(b)(i) electrophilic addition



(iii) 2-methyl-2-phenylbutanoic acid



Paper 3 Free Response Questions

Question 1 –

(a)(i) A weak acid is one that **dissociates partially** to **donate H⁺ ions**.

(ii) Since pH = 1.44

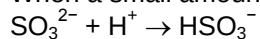
$$[H^+] = 10^{-1.44} = 3.63 \times 10^{-2} \text{ mol dm}^{-3}$$

$$K_{a1} = \frac{[H^+]^2}{c_{\text{acid}} - [H^+]} = \frac{(3.63 \times 10^{-2})^2}{0.20 - 3.63 \times 10^{-2}} = 8.05 \times 10^{-3} \text{ mol dm}^{-3}$$

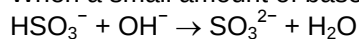
Or

$$K_{a1} = \frac{[H^+]^2}{c_{\text{acid}}} = \frac{(3.63 \times 10^{-2})^2}{0.20} = 6.59 \times 10^{-3} \text{ mol dm}^{-3}$$

(iii) When a small amount of acid is added,

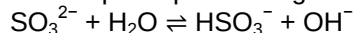


When a small amount of base is added,



(iv) The anion of the salt formed (at the second equivalence point), SO_3^{2-} , is a weak base, and undergoes hydrolysis in water to produce OH^- .

Hence, pH at point C is greater than 7.



(b) Both Oleum and sulfuric acid have simple molecular structure and they are held by hydrogen bonding between their respective molecules.
Oleum has much more electrons than sulfuric acid and hence has stronger instantaneous dipole – induced dipole (id-id) attraction than sulfuric acid.
re heat energy is required to overcome the stronger id-id attraction in oleum
⇒ higher melting point

(c)(i) Standard enthalpy change of neutralisation is the **heat change** when **one mole of water is formed** when an acid neutralises a base under standard conditions of **298 K and 1 atm**.



$$\text{Number of moles of methylthiol} = 1.00 \times 50 \times 10^{-3} = 0.0500 \text{ mol}$$

$$\text{Number of moles of NaOH} = 6.00 \times 100 \times 10^{-3} = 0.600 \text{ mol}$$

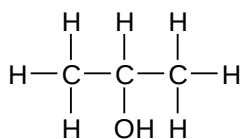
⇒ NaOH is in excess.

$$\text{Number of moles of H}_2\text{O produced} = 0.0500 \text{ mol}$$

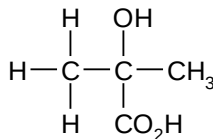
$$\text{Amount of heat evolved} = mc\Delta T = (150 \times 1.00) \times 4.20 \times 2.0 = 1260 \text{ J} = 1.26 \text{ kJ}$$

$$\text{Enthalpy change of neutralisation} = -\frac{1.26}{0.0500} = -25.2 \text{ kJ mol}^{-1}$$

(d)(i)



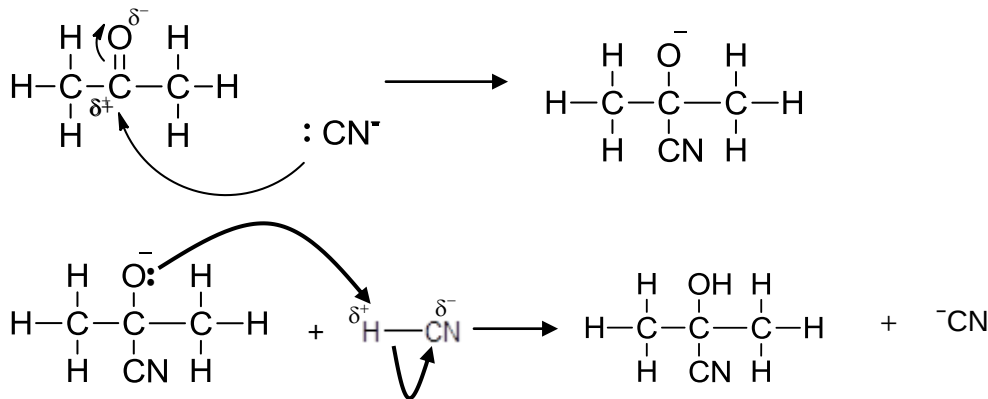
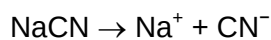
D



E

- (ii) Step I: cold, concentrated H_2SO_4 , followed by heating with H_2O
 Step II: KMnO_4 or $\text{K}_2\text{Cr}_2\text{O}_7$, $\text{H}_2\text{SO}_4(\text{aq})$, reflux
 Step IV: $\text{H}_2\text{SO}_4(\text{aq})$, reflux
 Step V: $\text{CH}_3\text{CH}_2\text{OH}$, concentrated H_2SO_4 , reflux

(iii) Nucleophilic Addition

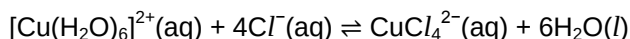


Question 2

- (a) Upon the initial addition of $\text{NH}_3(\text{aq})$ to the **(light) blue solution**, a **(light) blue ppt** will be formed. Upon addition of an excess of $\text{NH}_3(\text{aq})$, the **light blue ppt** dissolves to give a **dark blue solution**.
 (light) blue solution – $\text{Cu}^{2+}(\text{aq})$ or $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$
 (light) blue ppt – $\text{Cu}(\text{OH})_2(\text{s})$ or $\text{Cu}(\text{H}_2\text{O})_4(\text{OH})_2(\text{s})$
 dark blue solution – $[\text{Cu}(\text{NH}_3)_4]^{2+}(\text{aq})$ or $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}(\text{aq})$

- (b) (Since O^{2-} is acting as a *Bronsted* base, it will be able to abstract a proton from the solvent, in this case, NH_3).
 $\text{O}^{2-} + \text{NH}_3 \rightarrow \text{OH}^- + \text{NH}_2^-$ (or $\text{CuO} + 5\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]^{2+} + \text{OH}^- + \text{NH}_2^-$)
 The colour of the solution would be **dark blue** (due to the $[\text{Cu}(\text{NH}_3)_4]^{2+}$ complex)

- (c) The ligand exchange reaction is an equilibrium reaction:



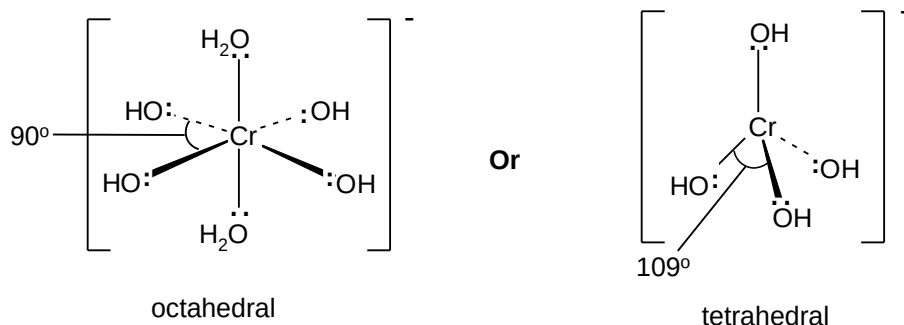
When concentrated hydrochloric acid is added, the equilibrium shifts to the right to produce the $\text{CuCl}_4^{2-}(\text{aq})$ ion, which is yellow in colour

(Since Cl^- is a weak ligand, the equilibrium position originally lies to the left).

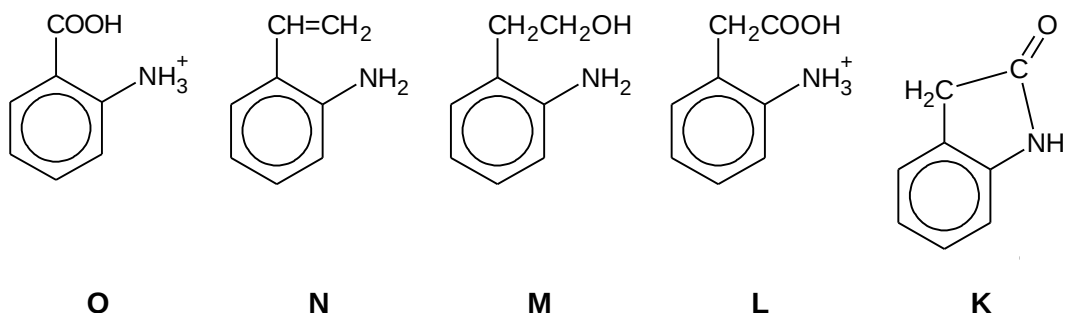
No change in colour when concentrated sulfuric acid was added because SO_4^{2-} is an even weaker ligand than Cl^- , and so $\text{Cu}^{2+}(\text{aq})$ does not undergo ligand exchange occurs with concentrated H_2SO_4 .

As water is added, the concentrated hydrochloric acid is diluted, and the equilibrium position shift back to the left to get back $\text{Cu}^{2+}(\text{aq})$ which is pale blue in colour.

- (d)



(e)

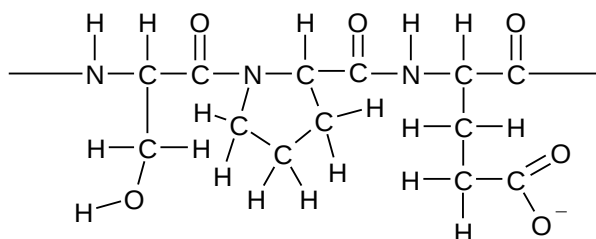


- 2-nitrobenzene undergoes **reduction** with Sn in concentrated HCl to give **O**, 2-aminobenzoic acid. The amino group is protonated under the acidic conditions.
- **N** is a **terminal alkene** as it undergoes **oxidative cleavage** to form a carboxylic acid **O** and CO₂ gas.
- **M** is an **alcohol** as it undergoes **elimination of H₂O** over hot Al₂O₃ to form an alkene **N**.
- **L** is a **carboxylic acid** as it undergoes **reduction** with LiAlH₄ to form a **primary alcohol M**.
- **K** is a **cyclic amide** as it undergoes **hydrolysis** to form only **L** which contains carboxylic acid and amine.

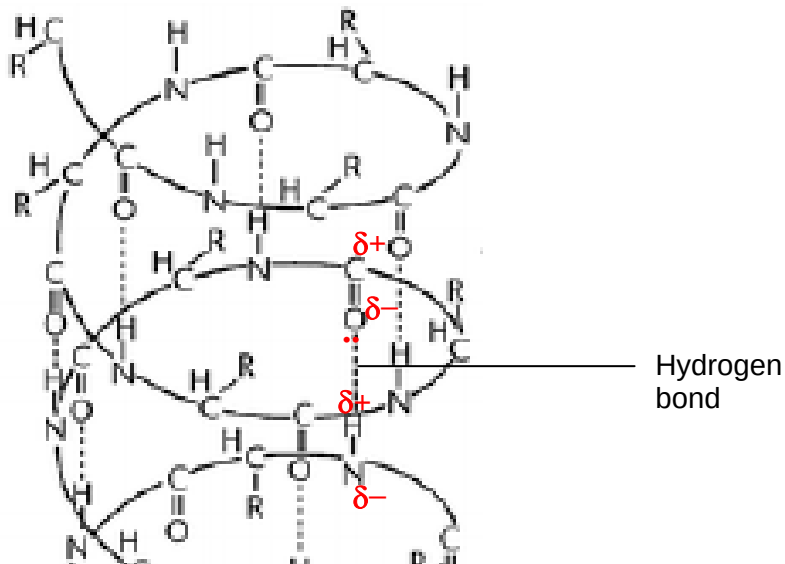
Question 3

(a)(i) asp and ser

(ii)



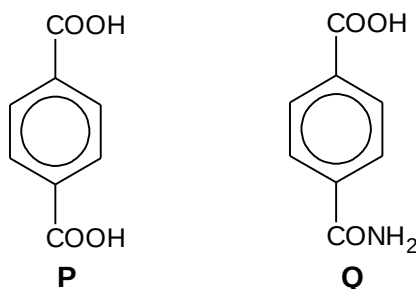
(iii)



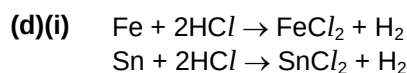
- (b)(i) The *quaternary structure* of a protein is the association of **more than one polypeptide chains** to form a **three-dimensional structure** that confers the biological function to the protein. The structure is stabilised by **instantaneous dipole - induced dipole interactions between non-polar R-groups, hydrogen bonds between polar uncharged R-groups, ionic interactions between oppositely charged R-groups and disulphide linkages between -SH groups of two cysteine residues**. Haemoglobin consists of 4 polypeptide chains, **2 α -subunits and 2 β -subunits** which are wound to give a spherical shape. Each polypeptide chain is **linked non-covalently** to a **haem group containing Fe²⁺ ion**.

- (ii) Valine contains a **non-polar alkyl group** that forms **instantaneous dipole - induced dipole interactions** while glutamic acid contains a **negatively charged -COO^- group (at pH = 7)** that forms **ionic interactions**.
The hydrophobic side-chain of valine is able to **interact with the non-polar patch on the surface of another haemoglobin molecule**, causing the haemoglobin molecules to aggregate.

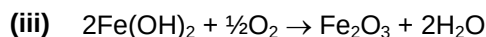
(c)(i)



- (ii) Step I: $(\text{CH}_3)_2\text{CHCl}$, anhydrous AlCl_3
Step IV: Concentrated NH_3 or $\text{NH}_3(\text{g})$, room temperature (in a sealed tube)
- (iii) Reagent: KMnO_4 with dilute H_2SO_4 and heat
Observations: Purple KMnO_4 decolourised with methylbenzene; KMnO_4 remains purple with PABA.
- Or**
- Reagent: $\text{Br}_2(\text{aq})$
Observations: Reddish brown Br_2 decolourised with methylbenzene; Br_2 remains reddish brown with PABA.
- Or**
- Reagent: $\text{Na} / \text{Na}_2\text{CO}_3$
Observations: Effervescence of CO_2/H_2 formed with PABA, which forms a white ppt with limewater/extinguishes a lighted splint with a 'pop' sound; no effervescence with methylbenzene.
- Or**
- Reagent: PCl_5 or SOCl_2 , anhydrous
Observations: White fumes of HCl formed with PABA; no white fumes with methylbenzene.



- (ii) To precipitate out Fe^{2+} as $\text{Fe}(\text{OH})_2(\text{s})$ - precipitate **R**

OrTo facilitate the separation of Fe^{2+} and Sn^{2+} (as Sn^{2+} will form a soluble complex in excess NaOH)

(iv)
$$n_{\text{Fe}_2\text{O}_3} = \frac{3.20}{2(55.8) + 3(16.0)} = \frac{3.20}{159.6} = 0.02005\text{mol}$$

$$\Rightarrow n_{\text{Fe}} = 2 \times n_{\text{Fe}_2\text{O}_3} = 2 \times 0.02005 = 0.0401\text{mol}$$

$$m_{\text{Fe}} = 0.0401 \times 55.8 = 2.24\text{g}$$

$$\Rightarrow m_{\text{Sn}} = 2.60 - 2.24 = 0.36\text{g}$$

$$\Rightarrow \% \text{ by mass of Sn} = \frac{0.36}{2.60} \times 100\% = 13.9\%$$

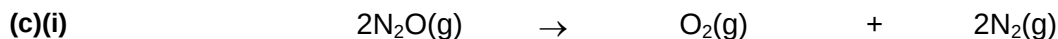
Question 4

- (a) Ionic radius: $\text{Mg}^{2+} = 0.065 \text{ nm}$ and $\text{Ba}^{2+} = 0.135 \text{ nm}$
 As the size of Mg^{2+} is smaller than Ba^{2+} , while the charge of the two cations are the same (+2), the charge density of Mg^{2+} is higher and hence its polarising power is stronger than that of Ba^{2+} .
 Mg^{2+} is therefore able to distort the electron cloud of NO_3^- to a greater extent and weakening the N–O bond in NO_3^- . Therefore, $\text{Mg}(\text{NO}_3)_2$ will decompose at a lower temperature than $\text{Ba}(\text{NO}_3)_2$.



(ii) $\Delta H_f^\ominus = [82 + 2(-242)] - (-366) = -36 \text{ kJ mol}^{-1}$

- (iii) For $\text{N}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{N}_2\text{O}(\text{g})$, entropy decreases (or $\Delta S < 0$) as there is a decrease in number of moles of gas (from $1\frac{1}{2}$ moles to 1 mole)
 Using $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$, since $\Delta H > 0$ (given in question) and $\Delta S < 0$,
 $\Rightarrow \Delta G > 0$ at all temperatures
 $\Rightarrow$ The reaction is not energetically feasible



$p_{\text{initial}} / \text{kPa}$	25.0		0		0
$p_{\text{change}} / \text{kPa}$	$-x$		$+\frac{1}{2}x$		$+x$
$p_{\text{total}} / \text{kPa}$	$25.0 - x$		$\frac{1}{2}x$		x

$$P_T = P_{\text{N}_2\text{O}} + P_{\text{N}_2} + P_{\text{O}_2}$$

$$\text{total pressure, } p = (25.0 - x) + \frac{1}{2}x + x = 25.0 + \frac{1}{2}x$$

$$x = 2p - 50$$

At any time, t ,

$$P_{\text{N}_2\text{O}} = P_T - P_{\text{O}_2} - P_{\text{N}_2}$$

$$= p - 0.5x - x$$

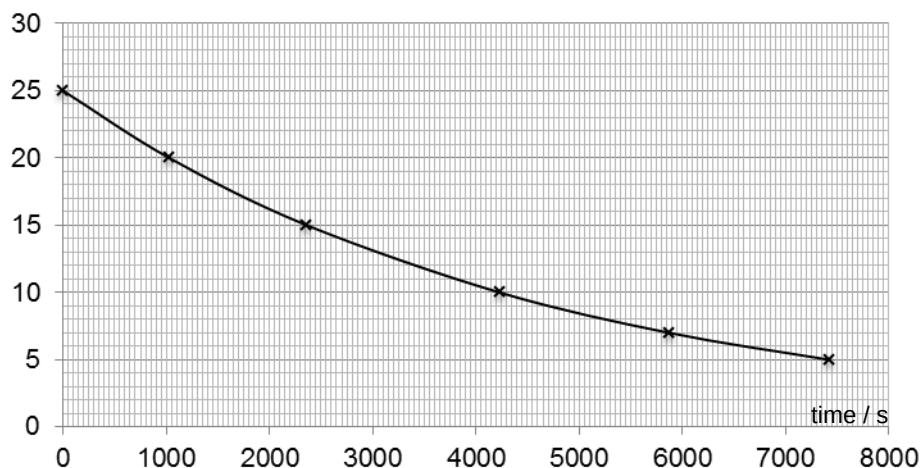
$$= p - 1.5x$$

$$= p - 1.5(2p - 50.0)$$

$$= 75.0 - 2p \text{ kPa}$$

(ii)

total pressure, p / kPa	25.0	27.5	30.0	32.5	34.0	35.0
time, t / s	0	1030	2360	4230	5870	7420
partial pressure of $\text{N}_2\text{O} / \text{kPa}$	25.0	20.0	15.0	10.0	7.0	5.0

(iii) partial pressure of N₂O / kPa

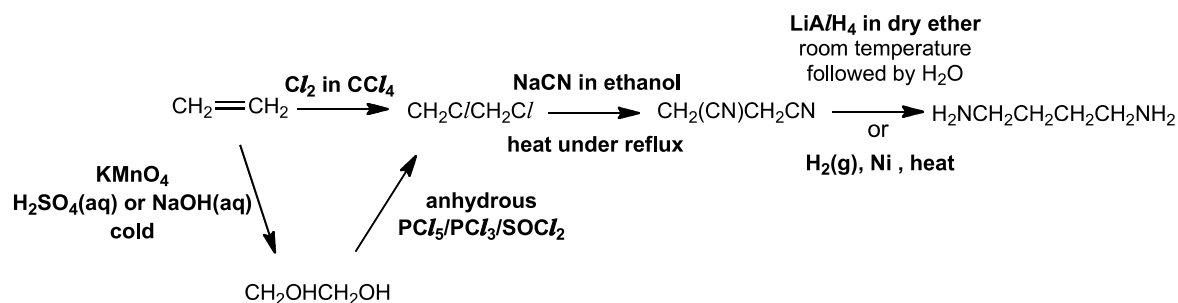
From the graph since 1st $t_{1/2}$ = 3200 s = 2nd $t_{1/2}$, the order of reaction with respect to N₂O is 1.
(must show 2 consecutive $t_{1/2}$ on the graph)

(iv) $k = \frac{\ln 2}{t_{1/2}} = 2.17 \times 10^{-4} \text{ s}^{-1}$

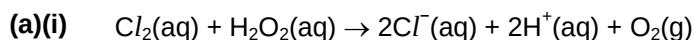
(d)(i) $^+\text{H}_3\text{N}-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2-\text{NH}_3^+$

The salt formed is ionic with strong electrostatic forces of attraction between the oppositely charged ions. The salt is thus non-volatile and does not vapourise easily and is odourless.

(ii)



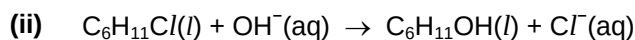
Question 5



Redox

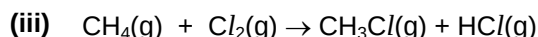
Iodine is not reduced by hydrogen peroxide because the $E^\ominus_{\text{cell}}$ is negative.

$$E^\ominus_{\text{cell}} = +0.54 + (-0.68) = -0.14 \text{ V}$$



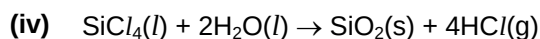
Nucleophilic Substitution

Chlorobenzene does not undergo nucleophilic substitution because the C-Cl has a partial double-bond character which makes it strong and more difficult to overcome.



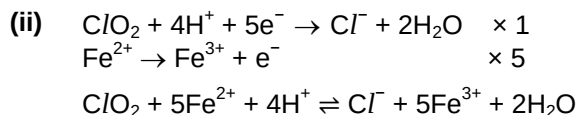
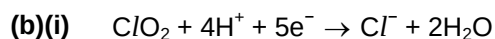
Free radical substitution

Methane does not undergo free radical substitution with iodine because the first step of the propagation stage is endothermic.



Hydrolysis

Carbon does not undergo the hydrolysis as it does not have available empty d-orbital to accept the lone pair of electrons from H₂O.



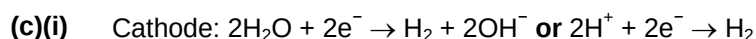
(iii) $0.73 = E^\ominus_{\text{reduction}} + (-0.77)$
 $E^\ominus_{\text{reduction}} = +1.50 \text{ V}$

(iv) Using $z = 5$,

$$\ln K_c = \frac{5 \times 96500 \times 0.73}{8.31 \times 298} = 142.23$$

$$K_c = e^{142.23} = 5.90 \times 10^{61} \text{ mol}^{-4} \text{ dm}^{12}$$

(v) As the value of the K_c is very large, the **equilibrium lies towards the right** and the **reaction is effectively complete**.



(ii) $Q = 0.30 \times (20 \times 60) = 360 \text{ C}$
 No. of moles of e^- transferred = $360 \div 96500 = 3.73 \times 10^{-3} \text{ mol}$
 No. of moles of OH^- liberated = $3.73 \times 10^{-3} \text{ mol}$
 $[\text{OH}^-] = (3.73 \times 10^{-3}) \div 0.450 = 8.29 \times 10^{-3} \text{ mol dm}^{-3}$
 $\text{pOH} = -\log(8.29 \times 10^{-3}) = 2.08$
 $\text{pH} = 14 - 2.08 = 11.9$

(d) High temperature

At high temperature, the gas particles possess high kinetic energy such that they have sufficient energy to overcome the intermolecular forces of attractions between them. Hence, the intermolecular forces between the gas particles are insignificant and negligible.

Low pressure

At low pressure, the gas particles are far apart from one another and so they have negligible volume as compared to the volume of the container. (In addition, the intermolecular attractions between the gas particles are also insignificant and negligible since they are very far apart.)

~ End of Suggested Answers ~